

Documentation Index — EMS Readiness Optimization

How to use this index: Each document has a status badge. Use  CURRENT docs for active reference.  HISTORICAL docs are preserved for project history and contain old metrics in context — don't use them as current reference.  REFERENCE docs are specialized analyses that should be cross-referenced with production results.

Quick Navigation

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Project overview	technical_report.md (technical_report.md) or executive_summary.md (executive_summary.md)
Which results files to use	data_usage_guide.md (data_usage_guide.md)
How to reproduce results	reproducibility_guide.md (reproducibility_guide.md)
Current metrics & numbers	optimization_results.md (optimization_results.md)
Math formulations	optimization_formulation.md (optimization_formulation.md)
Why P0 changed	nomenclature_migration.md (nomenclature_migration.md) ( historical)
Why capacity=2	capacity_sensitivity_analysis.md (capacity_sensitivity_analysis.md)
All project decisions	decisions_log.md (decisions_log.md)
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Figure traceability	figure_trace_guide.md (figure_trace_guide.md)
Run tests	testing_guide.md (testing_guide.md)

✓ CURRENT — Safe to Reference (34 docs)

Core Project Documentation

Document	Description
project_charter.md (project_charter.md)	Project scope, objectives, and problem statement
assumptions_log.md (assumptions_log.md)	All project assumptions with justification and risks
decisions_log.md (decisions_log.md)	Key decision records (DEC-001 through DEC-012)
blocker_log.md (blocker_log.md)	Obstacle tracking log
source_manifest.md (source_manifest.md)	Data source inventory and descriptions

Model Specifications

Document	Description
conceptual_model.md (conceptual_model.md)	Full conceptual model: entities, processes, assumptions
conceptual_model_selection.md (conceptual_model_selection.md)	Rationale for DES + MIP hybrid approach
demand_model_spec.md (demand_model_spec.md)	NHPP demand model specification
service_model_spec.md (service_model_spec.md)	Travel-time proxy and service-time distribution specs
optimization_formulation.md (optimization_formulation.md)	Mathematical formulations for P0/P1/P2 policies

Experimental Design & Results

Document	Description
experimental_design.md (experimental_design.md)	Experimental design: 5 experiment sets, 2,400 runs
optimization_results.md (optimization_results.md)	Optimization results summary (cap=2, current P0)
capacity_sensitivity_analysis.md (capacity_sensitivity_analysis.md)	Full-spectrum capacity sensitivity (cap 1-5)
cbd_robustness_analysis.md (cbd_robustness_analysis.md)	CBD robustness analysis with current metrics
cbd_comparison_and_validity_report.md (cbd_comparison_and_validity_report.md)	CBD comparison and model validity
cbd_definition.md (cbd_definition.md)	CBD geographic boundary definition
distance_metric_comparison.md (distance_metric_comparison.md)	Haversine vs Manhattan distance analysis
queue_analysis.md (queue_analysis.md)	Queue analysis confirming zero waiting

Reports & Summaries

Document	Description
technical_report.md (technical_report.md)	Full technical report (journal-style, ~600 lines)
executive_summary.md (executive_summary.md)	One-page executive summary
executive_presentation.md (executive_presentation.md)	Slide-style executive presentation
final_summary.md (final_summary.md)	Comprehensive project final summary
research_questions_assessment.md (research_questions_assessment.md)	Assessment against original research questions
project_alignment_verification.md (project_alignment_verification.md)	Alignment verification with project outline

Guides & References

Document	Description
data_usage_guide.md (data_usage_guide.md)	Which data/results files to use and why
notebook_guide.md (notebook_guide.md)	Notebook descriptions and execution order
reproducibility_guide.md (reproducibility_guide.md)	How to reproduce all results
testing_guide.md (testing_guide.md)	Testing framework and how to run tests
code_documentation.md (code_documentation.md)	Code architecture and module documentation
figure_trace_guide.md (figure_trace_guide.md)	Figure-to-source traceability guide
visualization_index.md (visualization_index.md)	Index of all project visualizations

Engineering & Quality

Document	Description
engineering_hygiene_summary.md (engineering_hygiene_summary.md)	Engineering quality practices summary
project_workflow_wbs.md (project_workflow_wbs.md)	Work breakdown structure
implementation_roadmap.md (implementation_roadmap.md)	12-month implementation roadmap



HISTORICAL — Preserved for Project History (18 docs)

⚠ These documents contain old metrics (e.g., P0 = 8.08 min, capacity = 5) **in historical context**. They document what changed and why. Do **not** use them as current reference for metrics or code behavior.

Policy & Nomenclature History

Document	Description
nomenclature_migration.md (nomenclature_migration.md)	P0 nomenclature migration history (old→new P0)
historical_data_policy.md (historical_data_policy.md)	Policy for handling historical vs current data
policy_tradeoff_analysis.md (policy_tradeoff_analysis.md)	Trade-off analysis using ORIGINAL index-based P0 (deprecated)
firehouse_capacity_analysis.md (firehouse_capacity_analysis.md)	Pre-DEC-010 capacity analysis (cap=5 era)

Audit & Migration Records

Document	Description
documentation_audit_report.md (documentation_audit_report.md)	Record of documentation audit and fixes applied
notebook_audit_report.md (notebook_audit_report.md)	Record of notebook audit and fixes applied
project_wide_audit_report.md (project_wide_audit_report.md)	Comprehensive project-wide audit record
honest_audit_assessment.md (honest_audit_assessment.md)	Honest assessment of project state
gap_analysis_report.md (gap_analysis_report.md)	Gap analysis and identified issues
gap_closure_report.md (gap_closure_report.md)	Gap closure progress report
gap_remediation_plan.md (gap_remediation_plan.md)	Remediation plan for identified gaps
fix_summary_4_issues.md (fix_summary_4_issues.md)	Summary of 4 critical fixes (cap=5→2, etc.)
filename_mismatch_fix_report.md (filename_mismatch_fix_report.md)	Filename mismatch fixes
output_comparison_report.md (output_comparison_report.md)	Before/after comparison of regenerated outputs
migration_plan.md (migration_plan.md)	Data and results migration plan

Phase History

Document	Description
phase_comparison_summary.md (phase_comparison_summary.md)	Phase-by-phase comparison summary
phase21_assessment.md (phase21_assessment.md)	Phase 2.1 assessment
final_status.md (final_status.md)	Status snapshot at a point in time



REFERENCE — Specialized Analysis (10 docs)

These are supplementary analysis documents. They are current but specialized — cross-reference with production results in `results/production_v2/`.

Specialized Analysis

Document	Description
<code>alternative_analyses_summary.md</code> (alternative_analyses_summary.md)	Summary of CBD/distance alternative analyses
<code>analysis_precinct_demand_trace.md</code> (analysis_precinct_demand_trace.md)	Precinct-level demand analysis trace
<code>cbd_focused_optimization_analysis.md</code> (cbd_focused_optimization_analysis.md)	CBD-focused optimization deep-dive
<code>fleet_sensitivity_dual_investigation.md</code> (fleet_sensitivity_dual_investigation.md)	Fleet sensitivity dual investigation
<code>output_analysis.md</code> (output_analysis.md)	Output analysis methodology
<code>eda_and_data_split_summary.md</code> (eda_and_data_split_summary.md)	EDA and data split summary
<code>verification_log.md</code> (verification_log.md)	V&V verification log

Project Management

Document	Description
<code>data_dependency_analysis.md</code> (data_dependency_analysis.md)	Data dependency and lineage analysis
<code>file_inventory.md</code> (file_inventory.md)	Complete file inventory
<code>project_archive.md</code> (project_archive.md)	Project archive structure

PDF Versions

Most `.md` files have corresponding `.pdf` versions (60 PDFs total). PDFs are auto-generated from Markdown and share the same status as their source `.md` file.

Current Key Metrics (for reference)

Metric	Value	Source
P0 mean RT (K=20)	3.17 min	Spatially-stratified baseline
P2 mean RT (K=20)	2.57 min	Demand-weighted MIP
8-min coverage (P2, K=20)	99.6%	Production simulation
Default capacity	2 units/firehouse	DEC-010
P0 nomenclature	Spatially-stratified	DEC-011/DEC-012
Total production runs	2,400	5 experiment sets × 30 rep- lications

Last audited: March 20, 2026 — 62 .md files, 60 .pdf files